

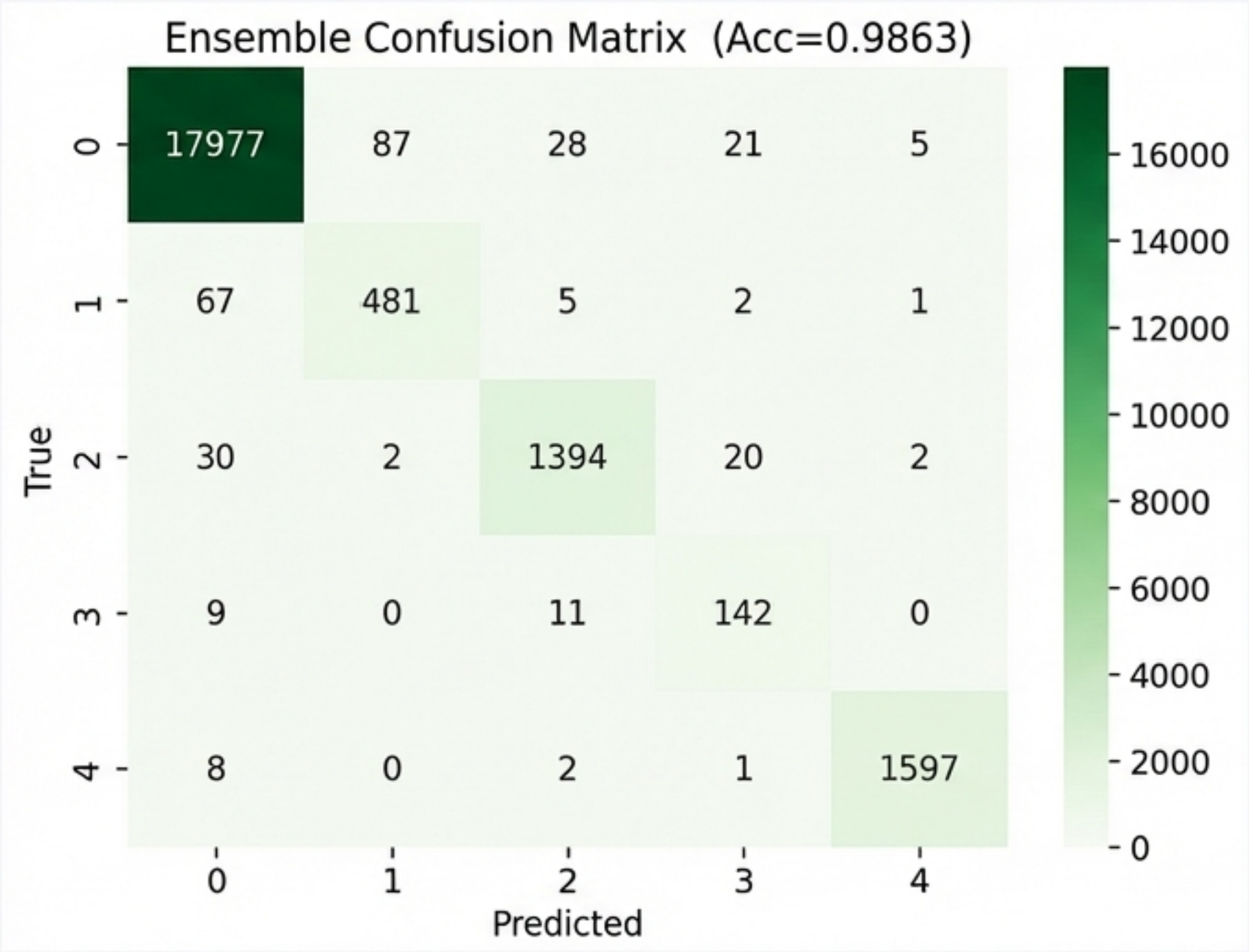
Deep learning for end-to-end classification of cardiac arrhythmias

A comparative analysis of 4 architectures (CNN, ResNet, BiLSTM, Transformer) with robustness validation and transfer learning.

MIT-BIH accuracy: 98.63%

Ensemble macro-F1: 0.9246

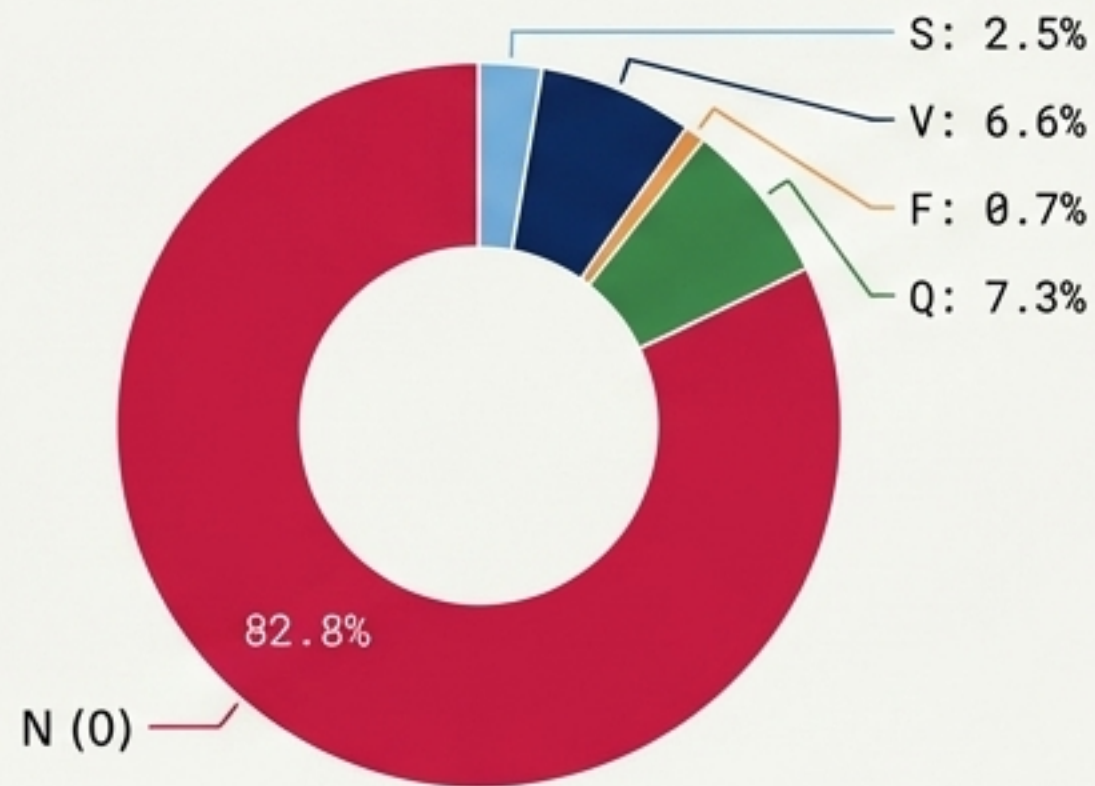
PTB transfer accuracy: 96.87%



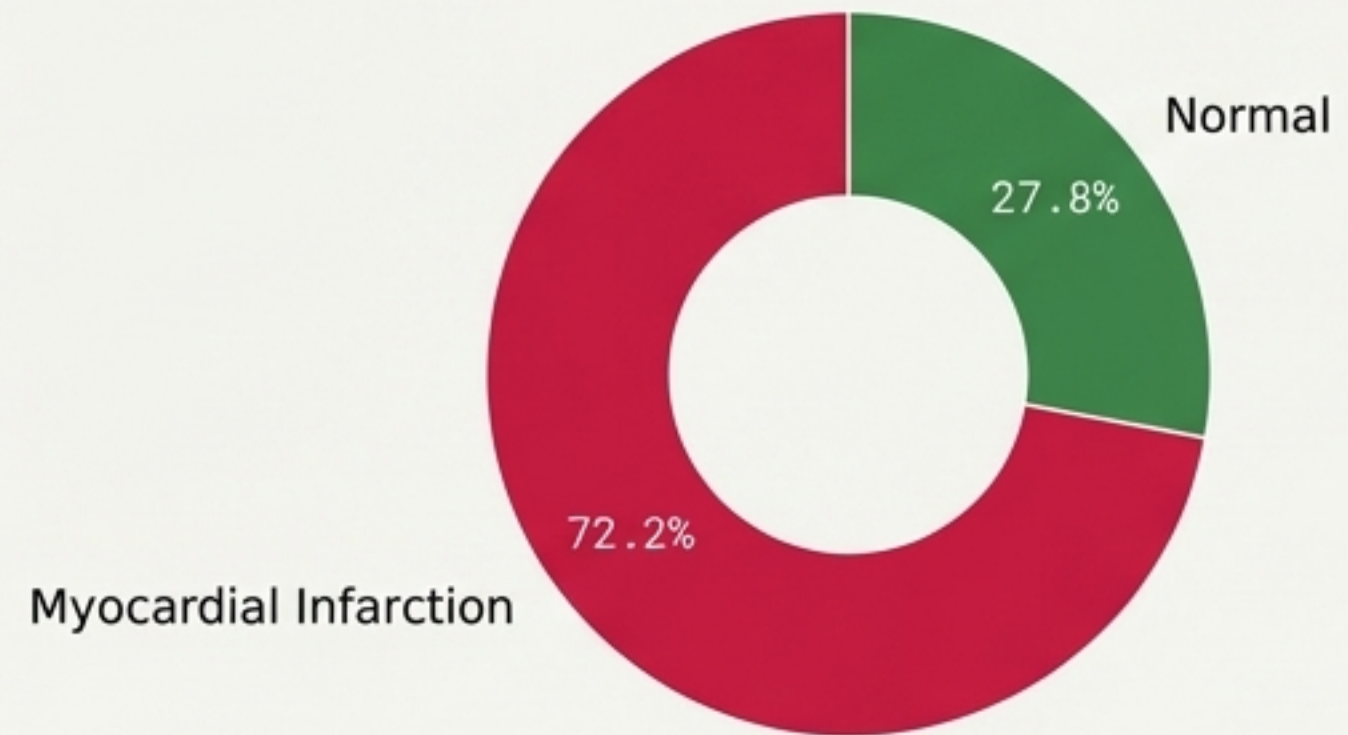
The data domain: extreme clinical imbalance

Training on real clinical data requires managing severe imbalance. The MIT-BIH dataset (AAMI multiclass classification) is 82.8% dominated by normal beats, obscuring rare but critical arrhythmias. The PTB dataset (infarction diagnosis) serves as a generalization test.

MIT-BIH Train — Class distribution

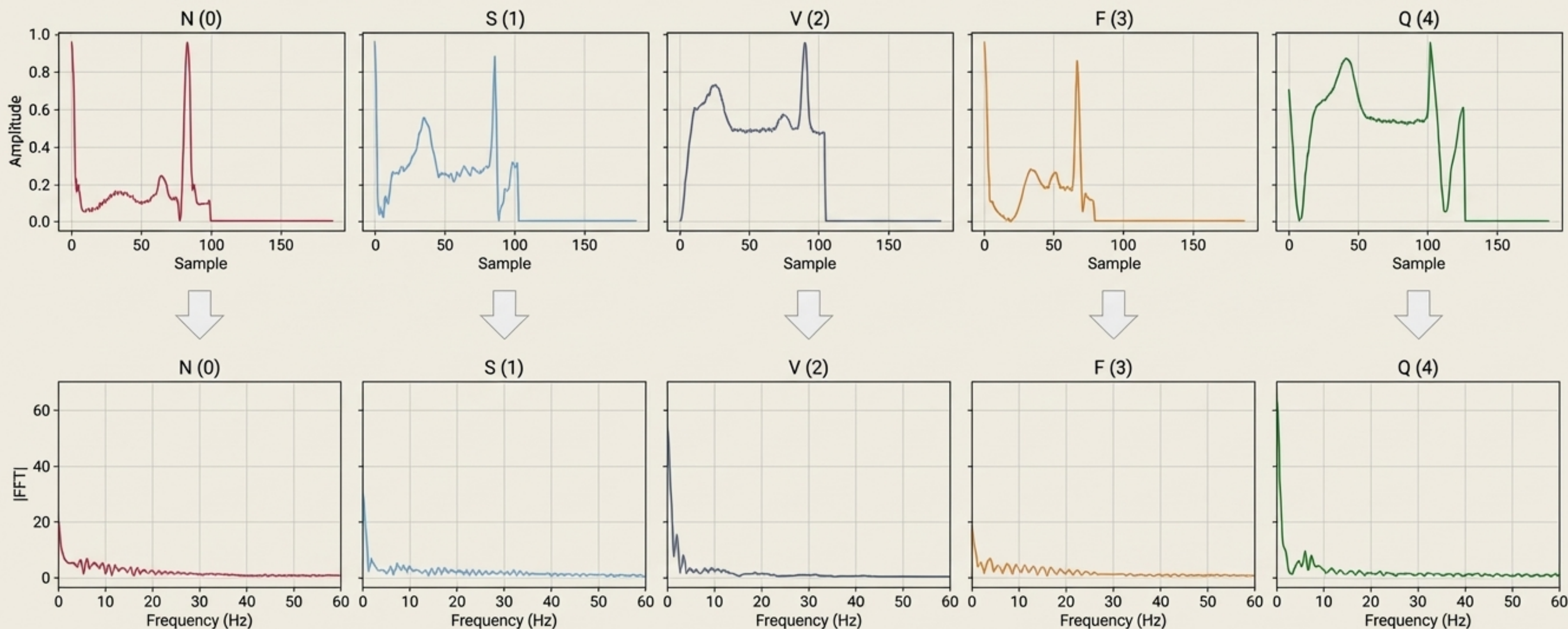


PTB — Class distribution



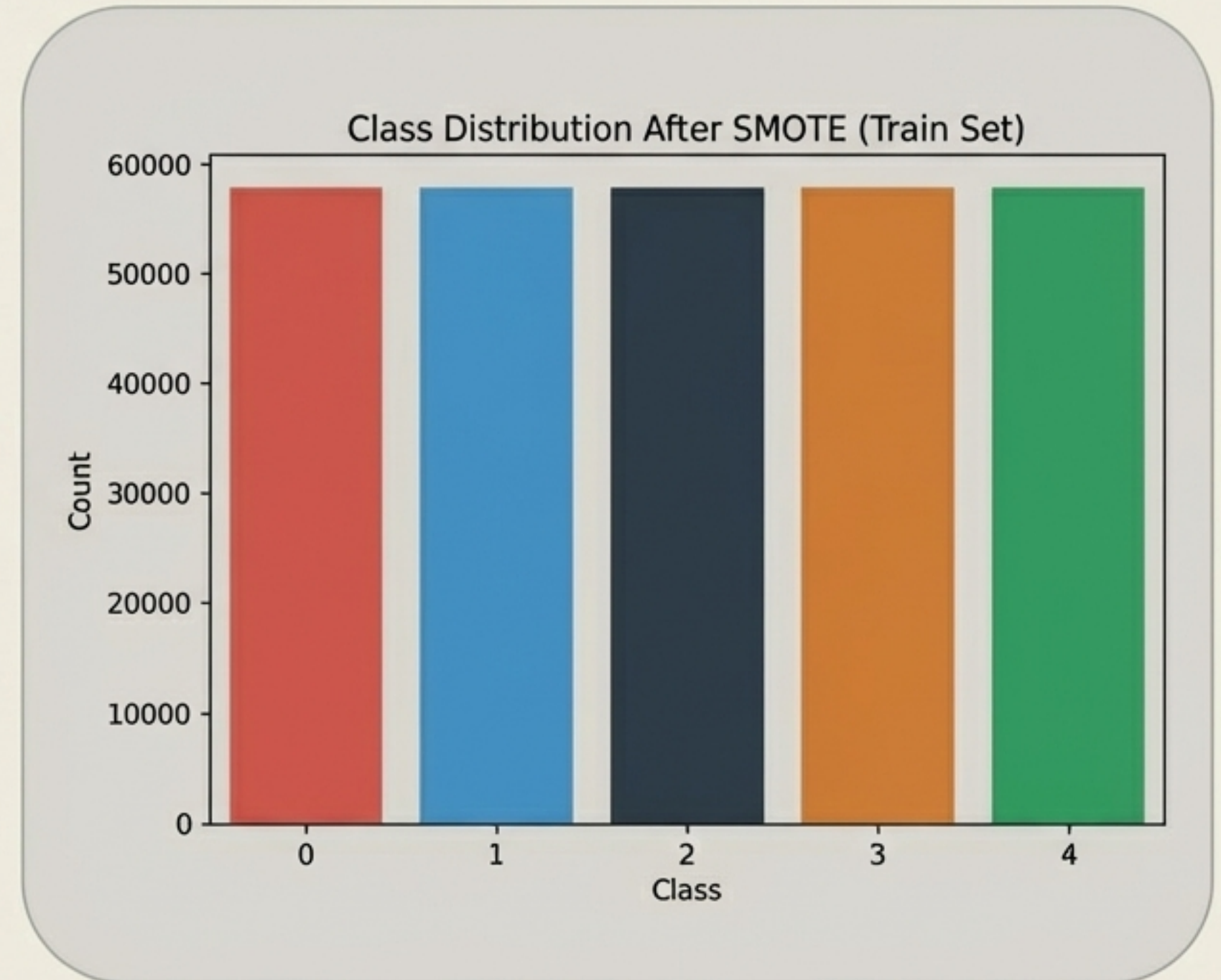
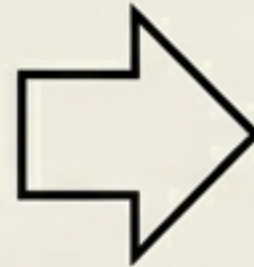
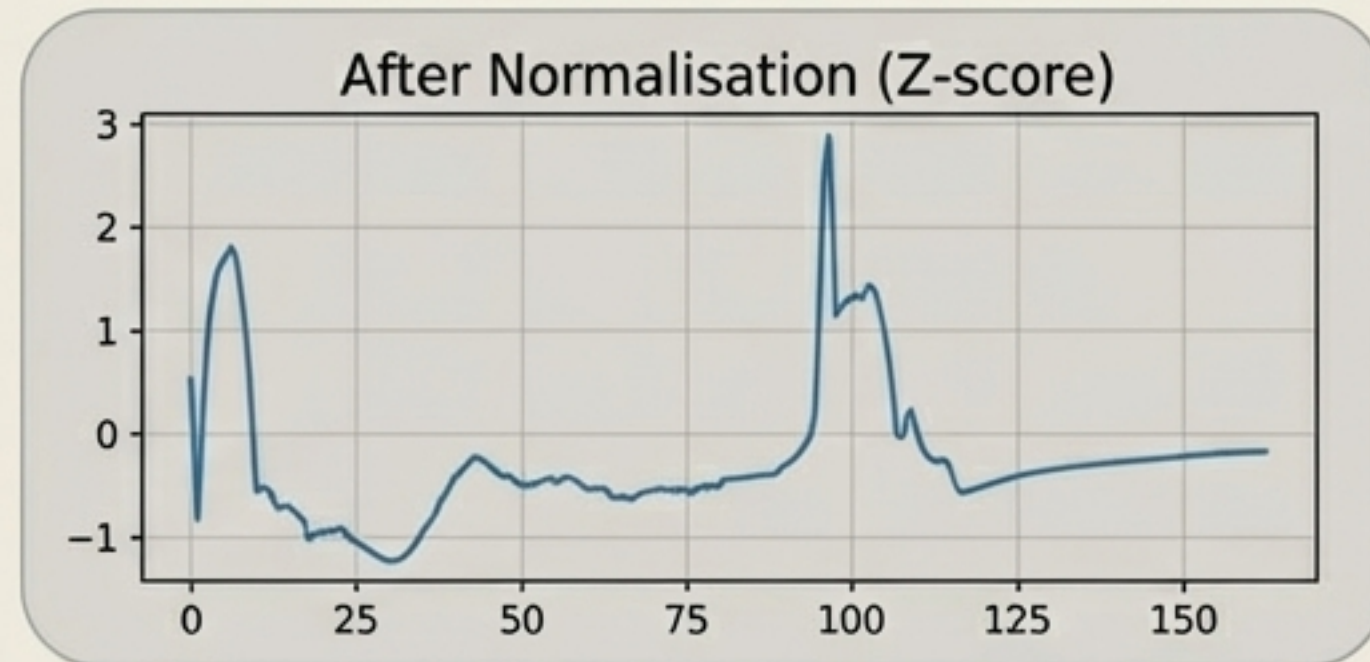
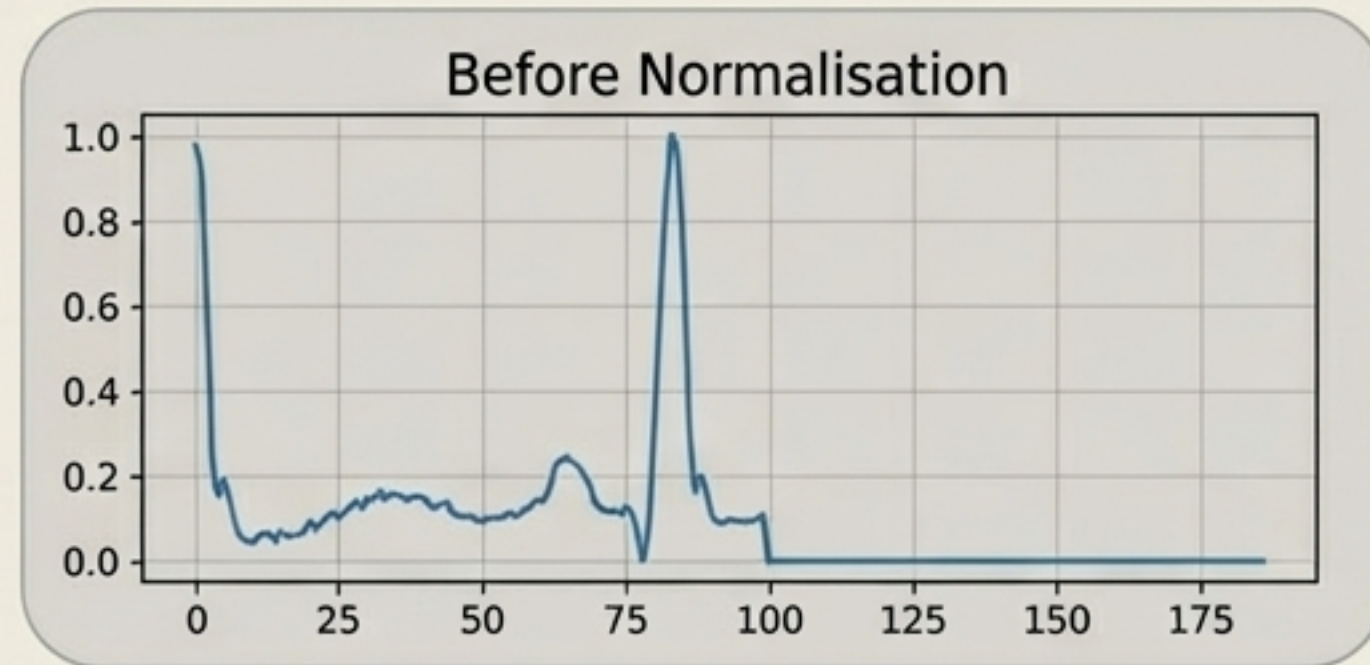
Signal analysis: morphology and frequency spectrum

Morphological differences between ectopic and normal beats are evident in the time domain, but FFT analysis reveals unique spectral signatures in the frequency domain, which are fundamental for 1D convolution-based neural networks.



Preprocessing pipeline: normalization and oversampling

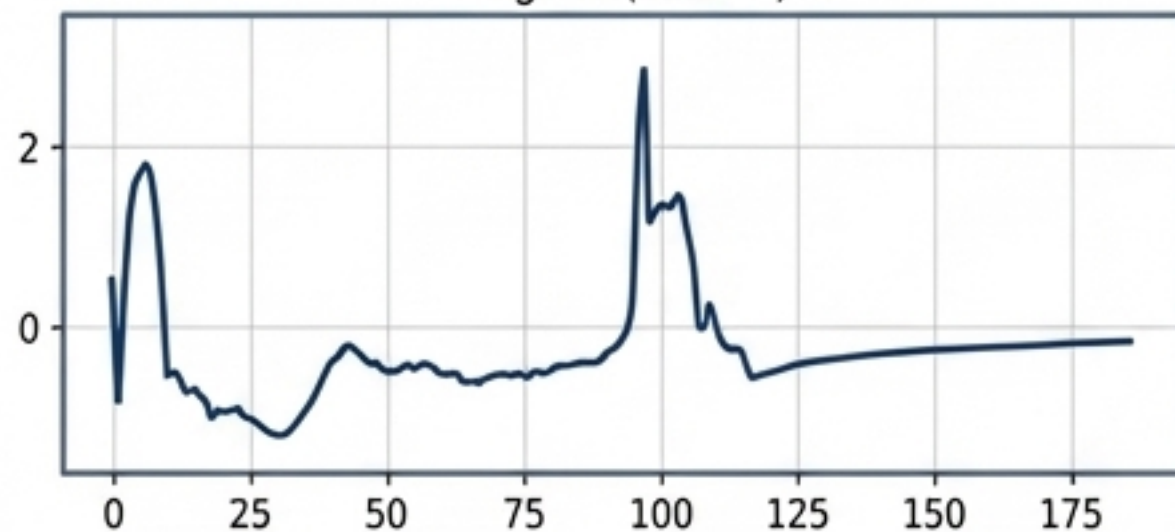
Raw signals exhibit high amplitude variance across patients. By applying Z-score normalization and SMOTE (Synthetic Minority Oversampling), minority classes are synthesized up to 72,000 samples each, preventing majority class bias.



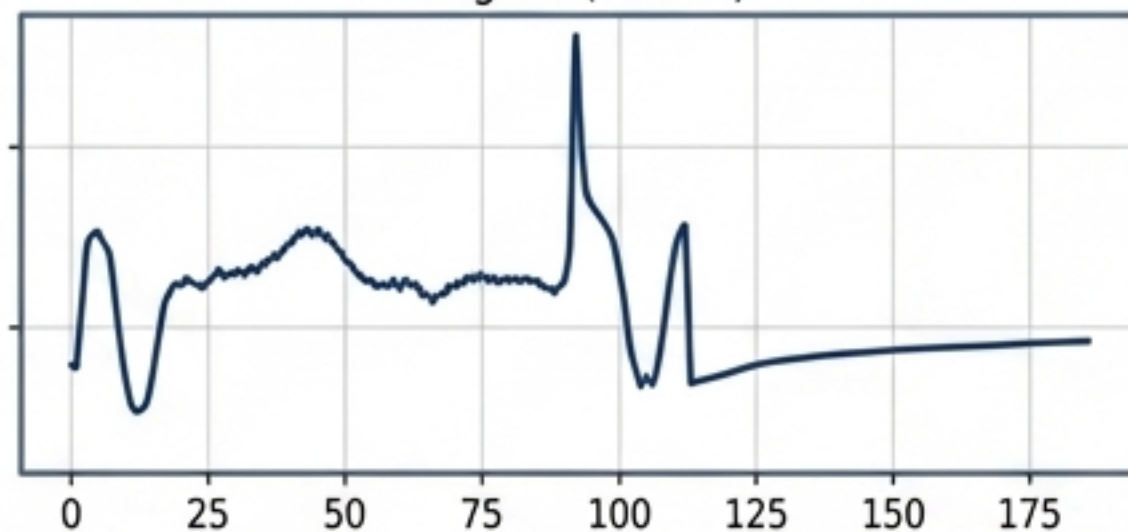
Dynamic data augmentation for sensor robustness

To simulate real-world ECG sensor noise and prevent overfitting, a custom generator applies physical perturbations on-the-fly: Gaussian noise, baseline wander (low-frequency sinusoids), amplitude scaling, and nonlinear time warping.

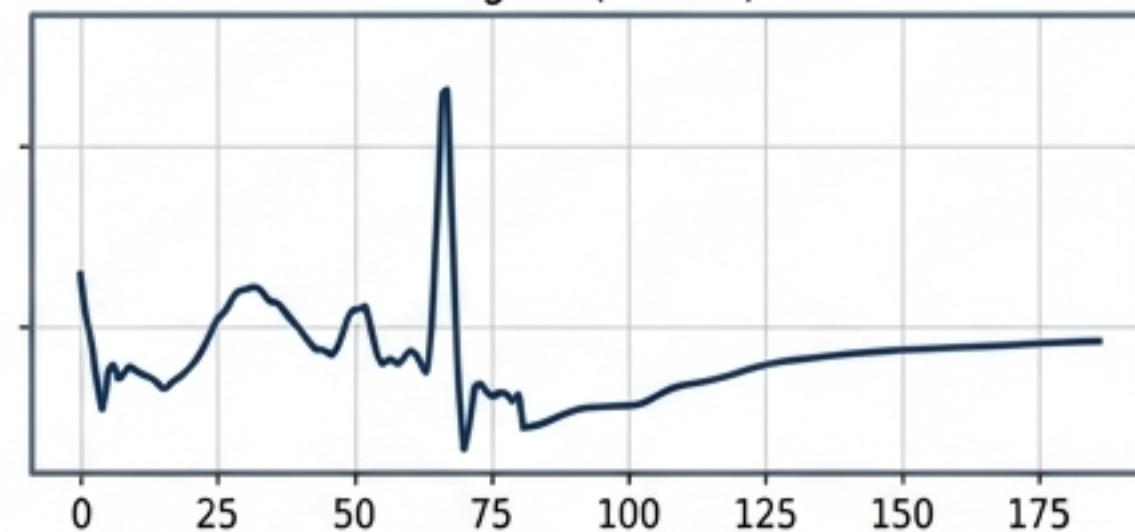
Original (class 4)



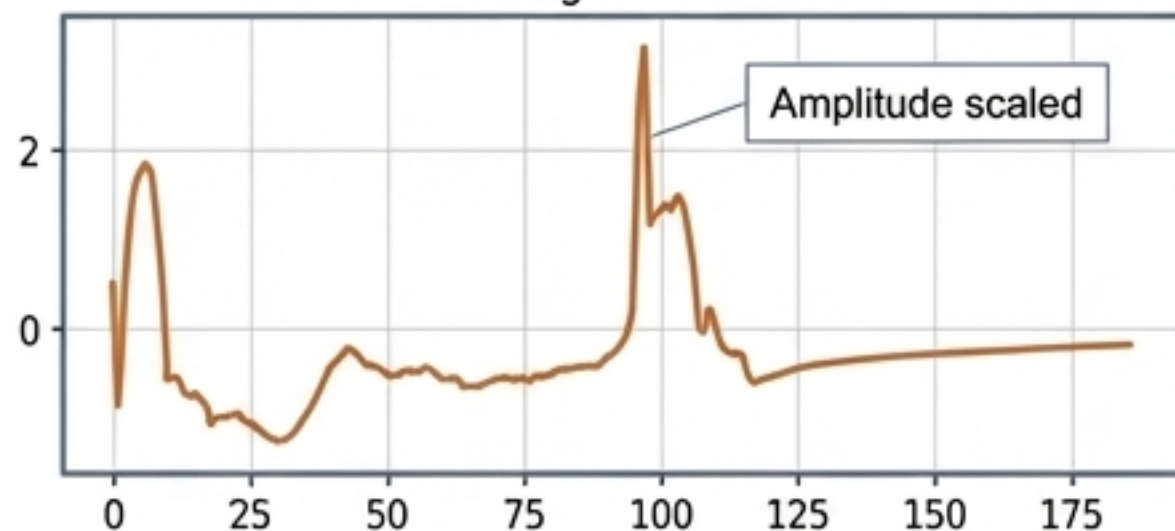
Original (class 4)



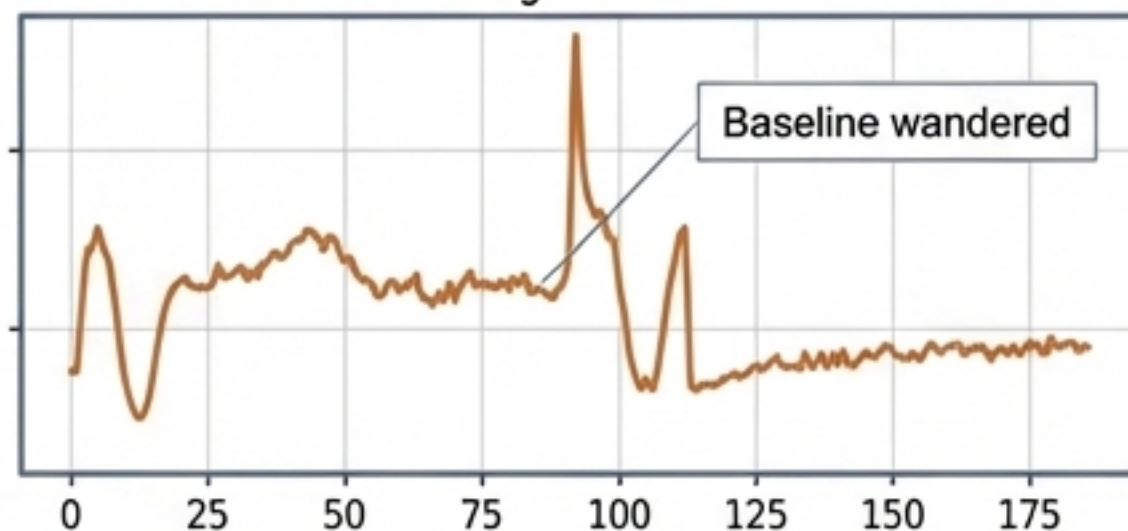
Original (class 3)



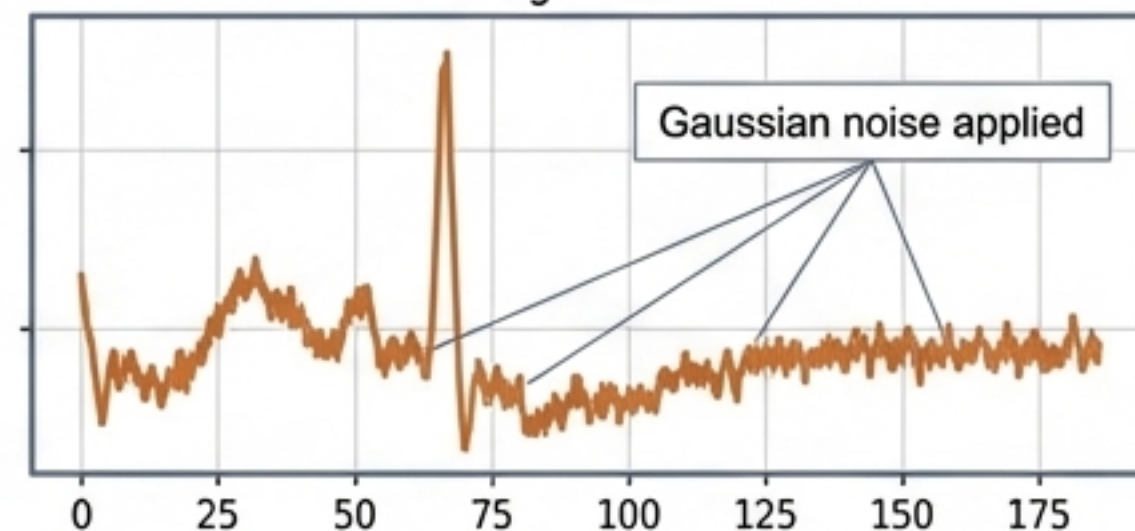
Augmented



Augmented

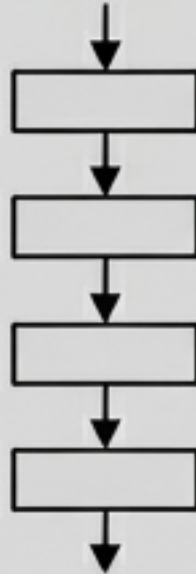
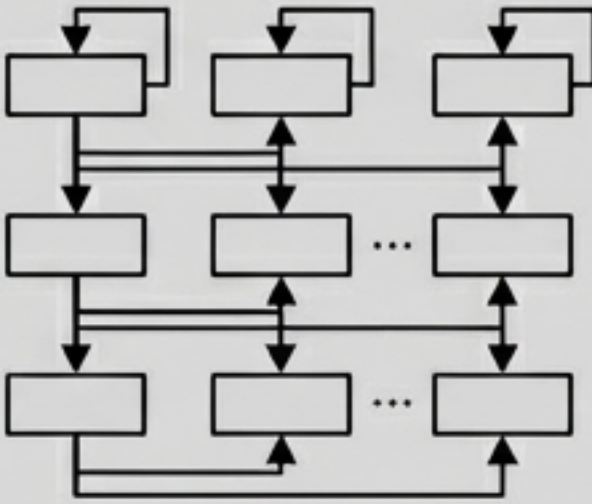
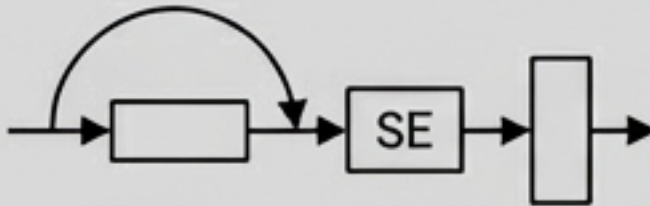
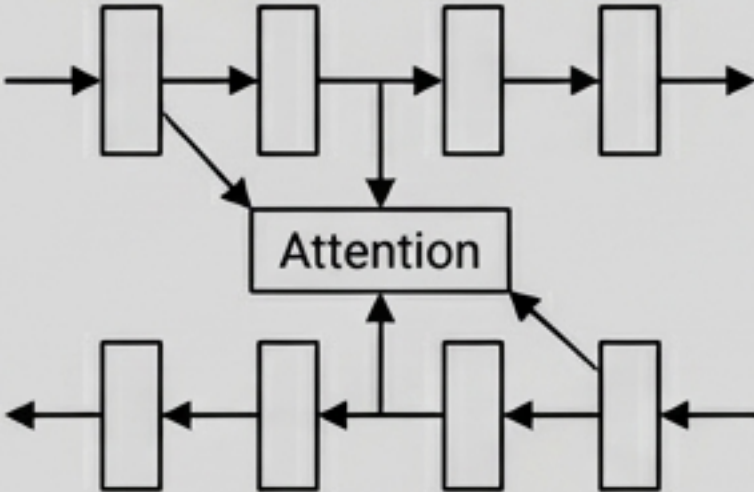


Augmented



Synthesis matrix: comparing 4 architectures

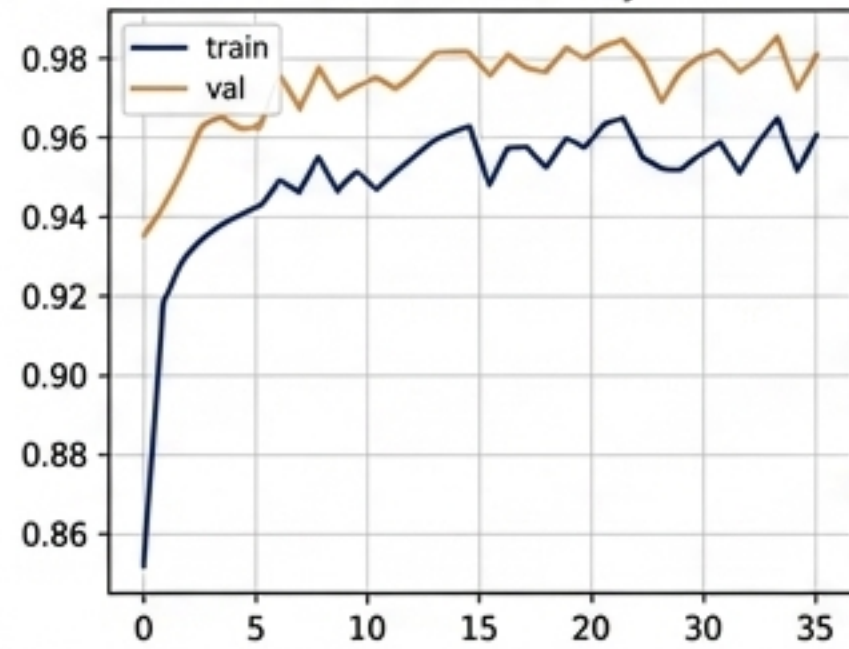
Architectures selected to evaluate the trade-off between purely spatial feature extraction (CNN), temporal abstraction (BiLSTM), and global attention (Transformer).

	CNN Baseline	Transformer1D	SE-ResNet	BiLSTM + Attention
Key mechanism	1-D standard convolutions	Multi-head self-attention with patch embeddings	Residual blocks + Squeeze-and-Excitation	Bidirectional temporal dependencies with learned soft-attention
Parameters	53k	106k	240k	371k
Network Topology				

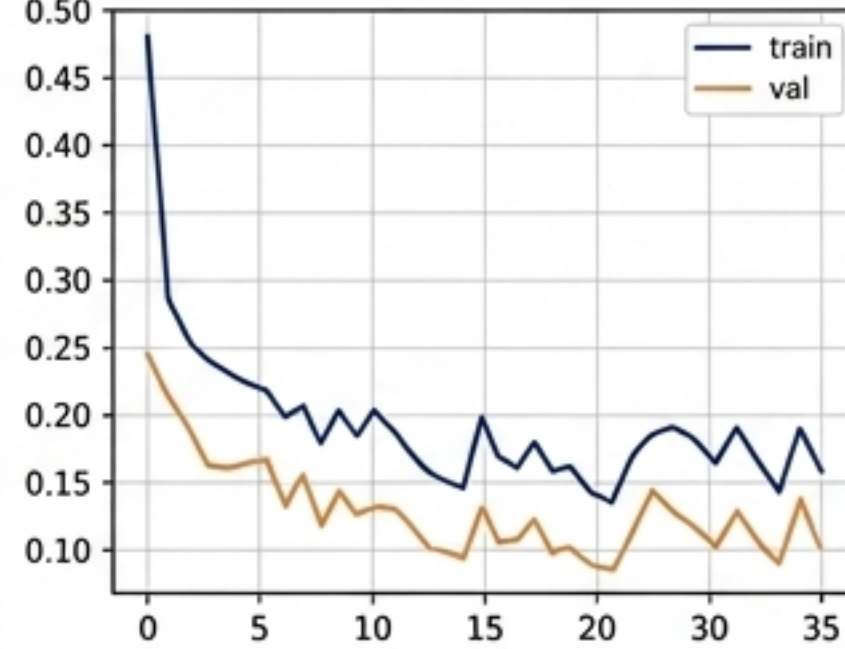
Training dynamics and convergence

Adam optimizer with categorical cross-entropy and cosine-annealing learning rate. SE-ResNet and Transformer show exceptionally stable convergence, validating the effectiveness of dynamic data augmentation.

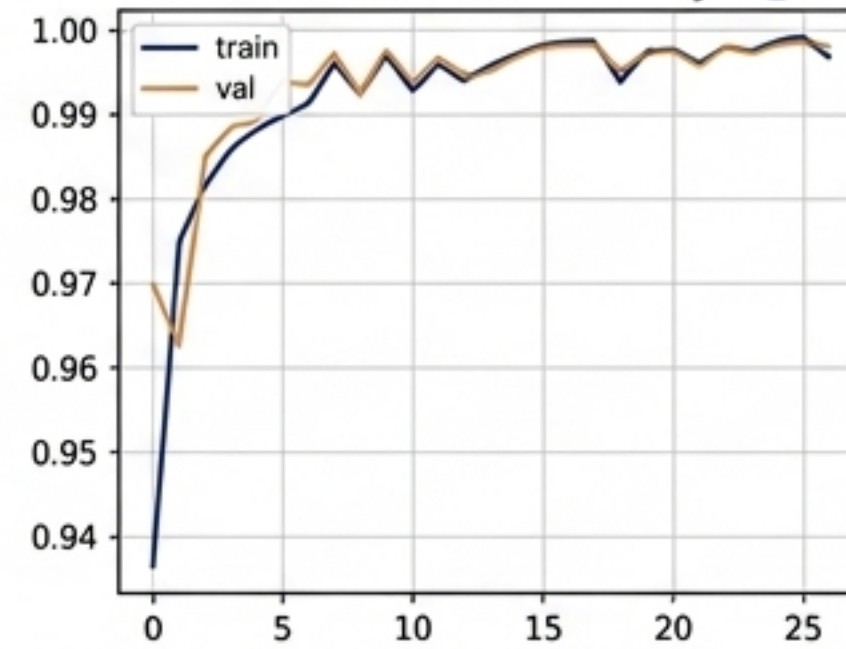
CNN – Accuracy



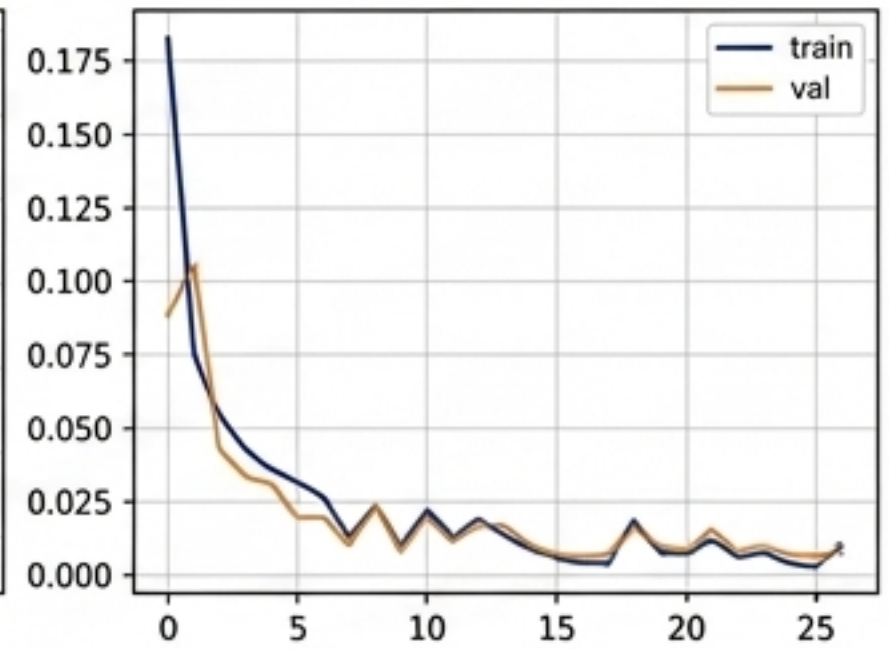
CNN – Loss



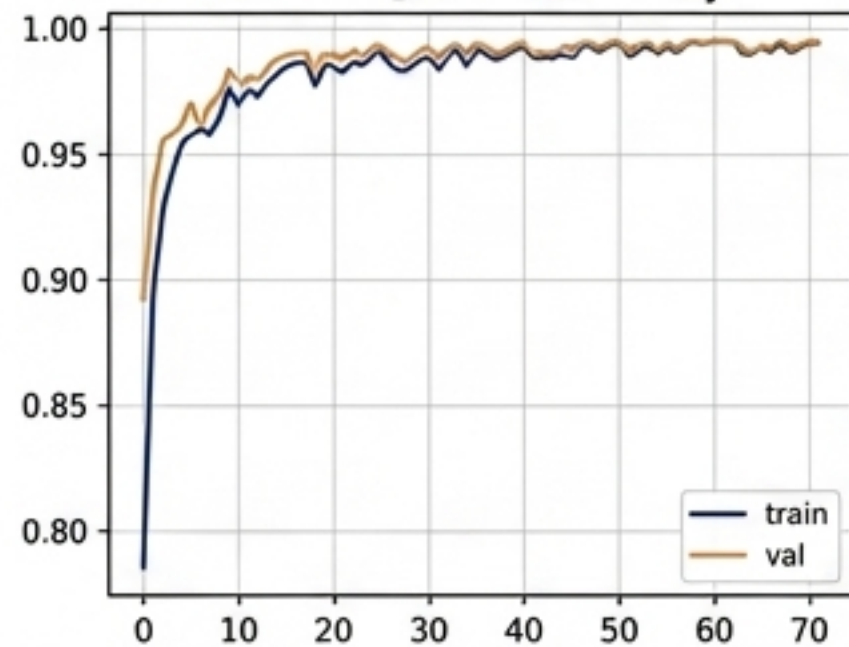
SE-ResNet – Accuracy



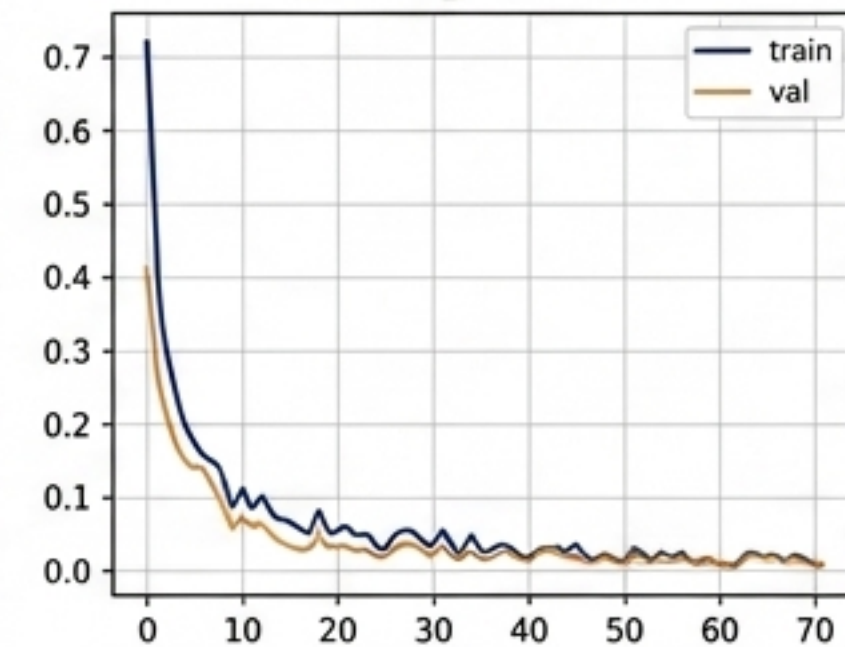
SE-ResNet – Loss



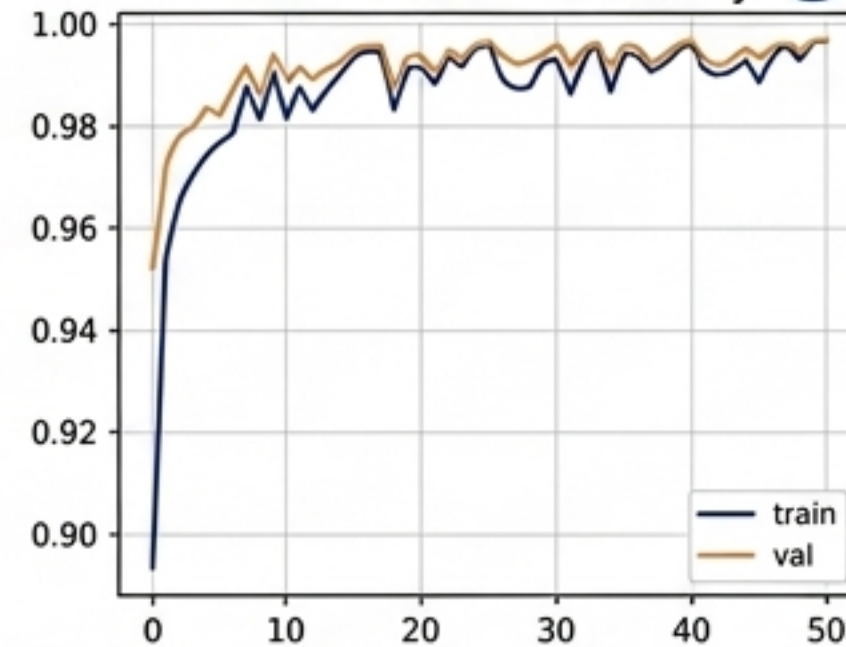
BiLSTM_Attn – Accuracy



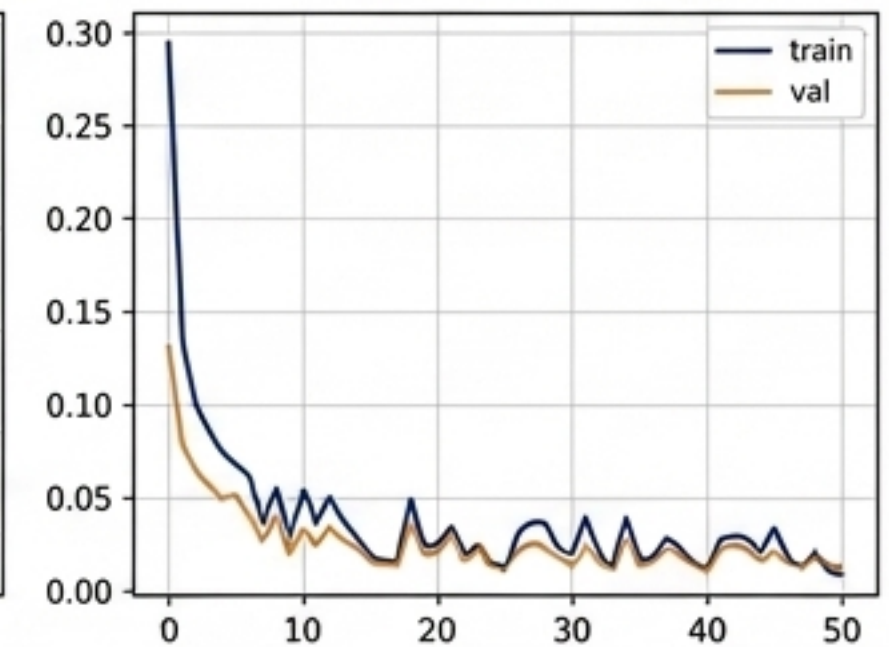
BiLSTM_Attn – Loss



Transformer1D – Accuracy

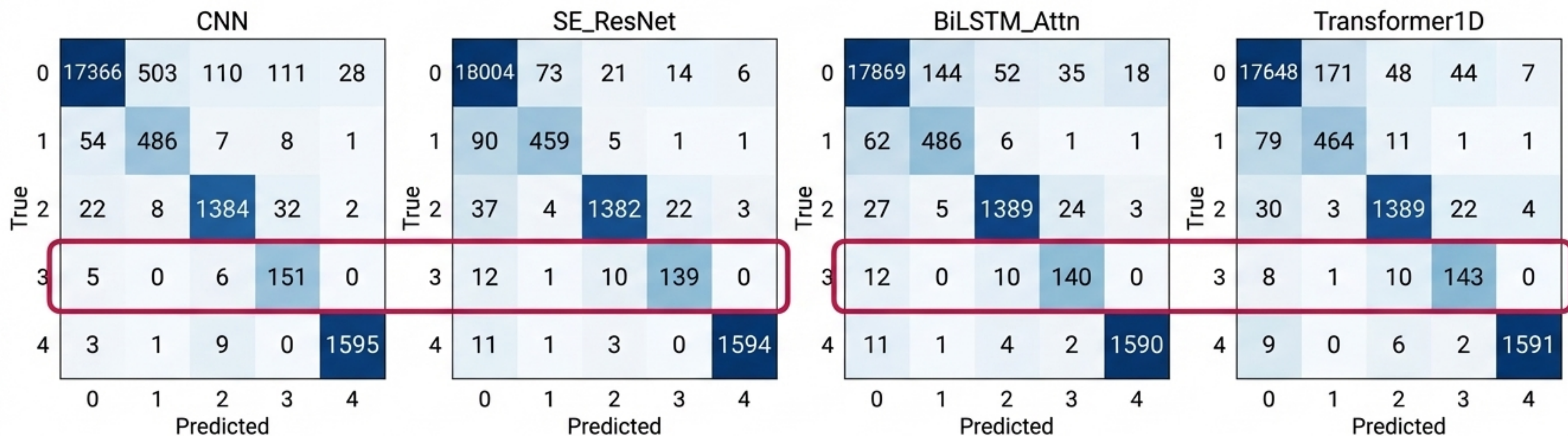


Transformer1D – Loss



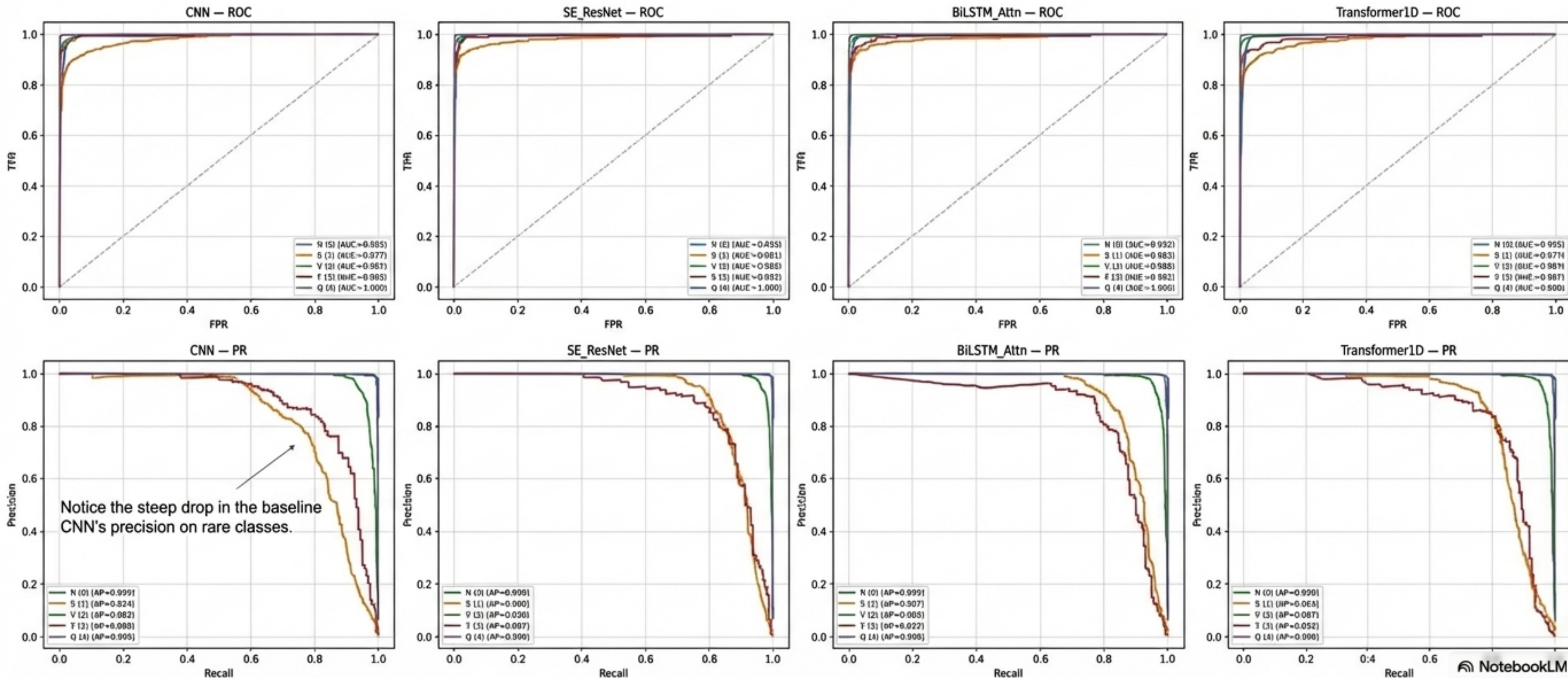
Raw performance evaluation on the test set

While the baseline CNN easily confuses class F (Fusion beats), the attention mechanisms and residual blocks (Transformer and SE-ResNet) resolve morphological overlap, ensuring superior recall.



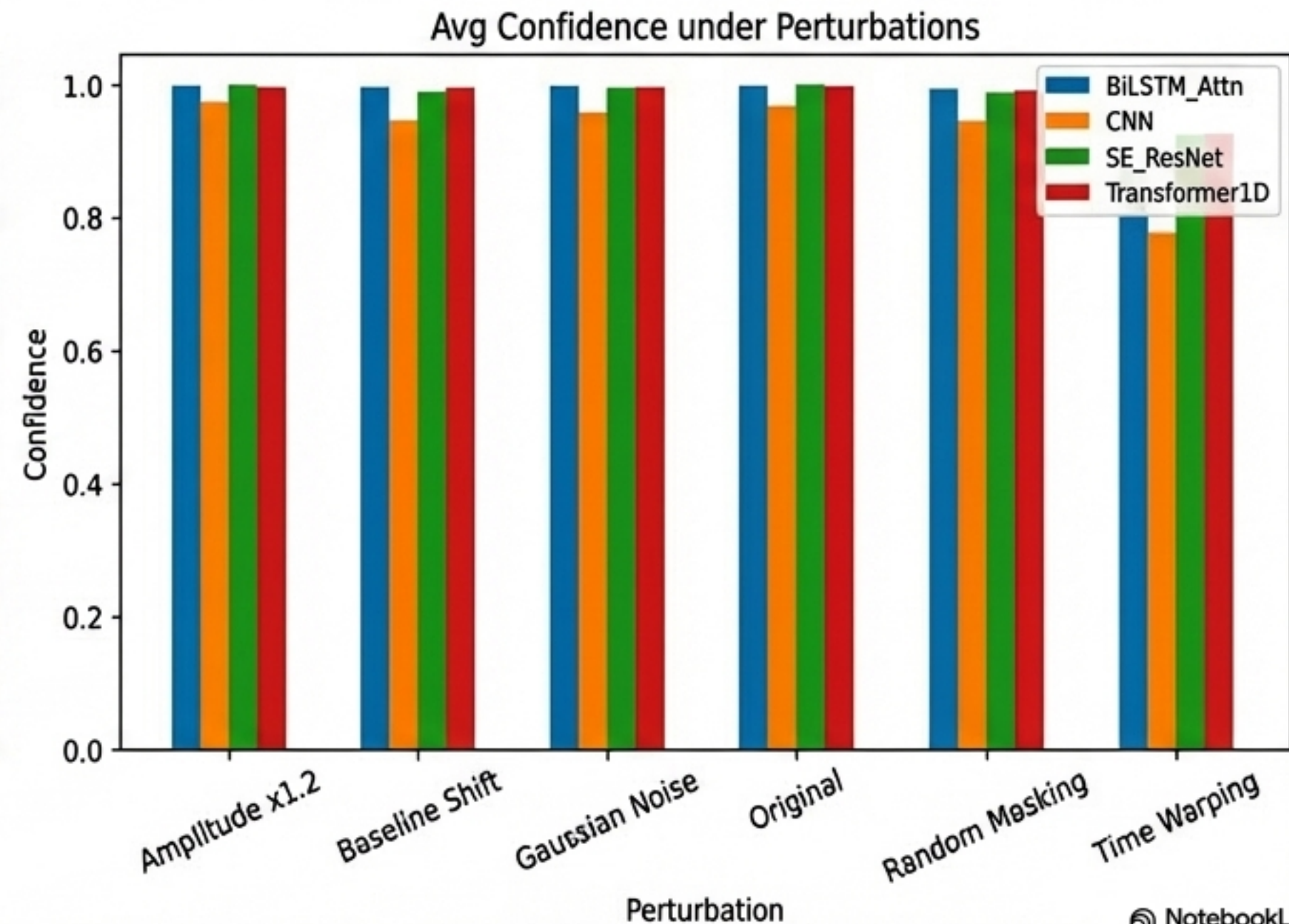
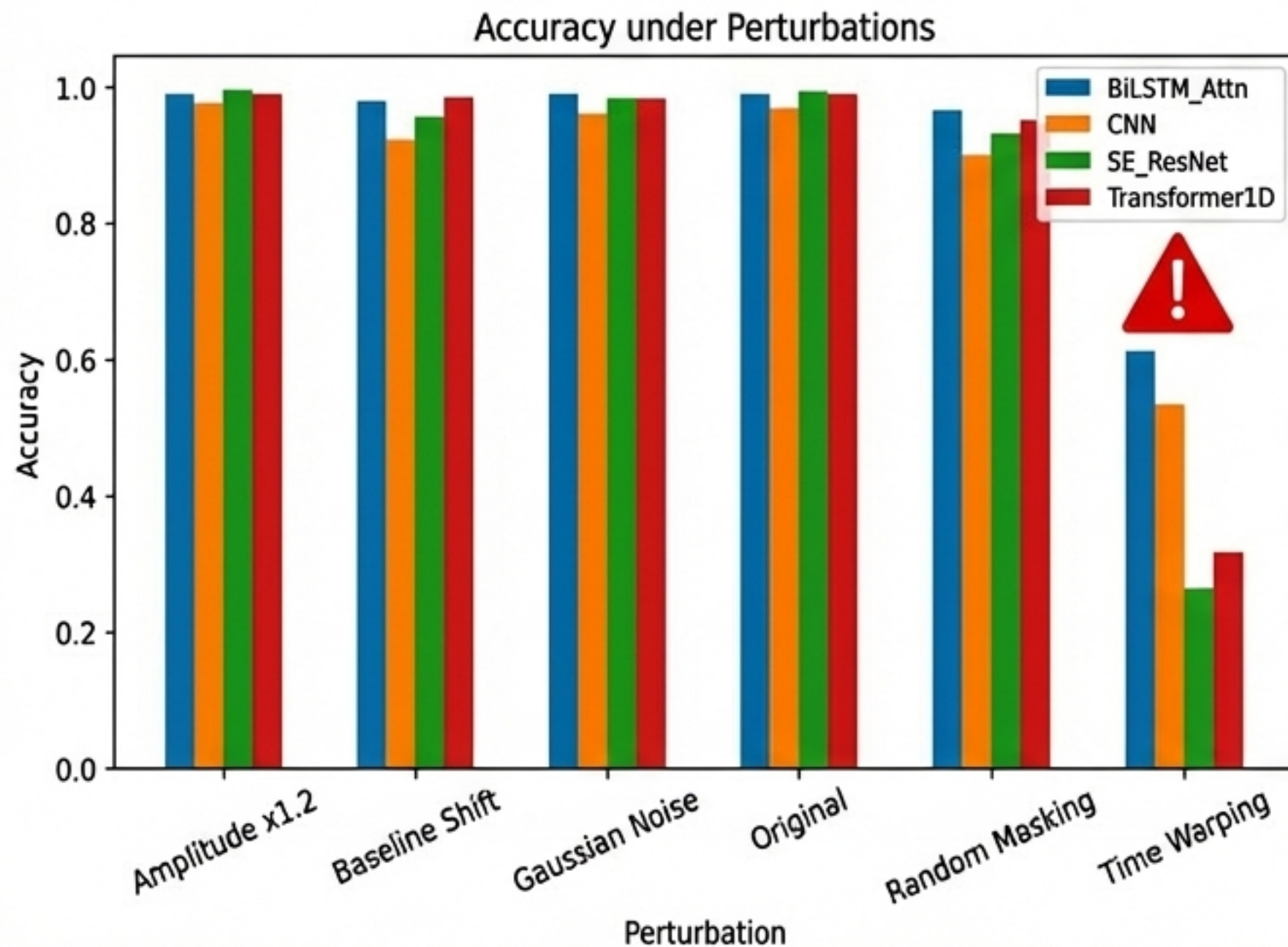
Metrics for imbalance: ROC vs Precision-Recall

For highly imbalanced medical datasets, the Area Under Curve (AUC) of ROC curves appears nearly perfect for all models. Precision-Recall curves, however, reveal the true capacity of advanced models to maintain high precision on minority classes.



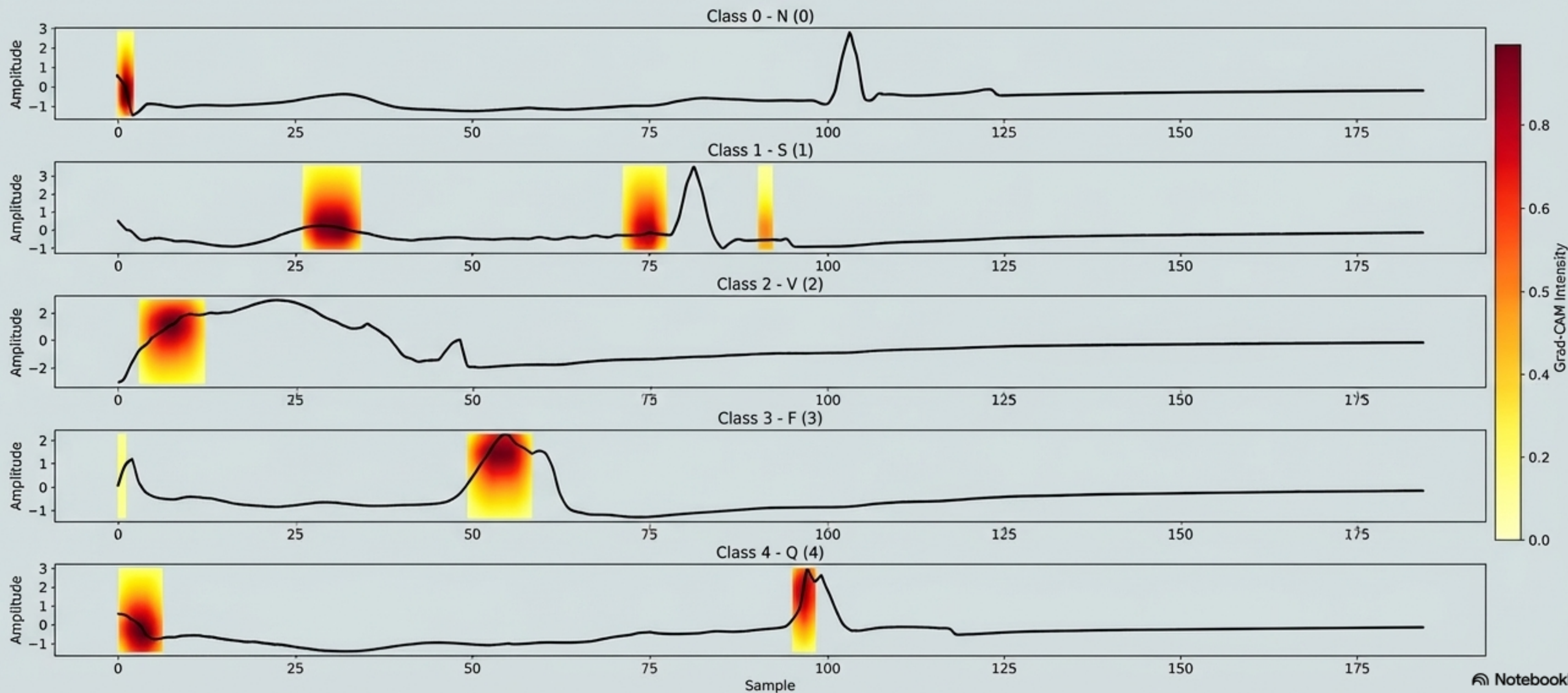
Stress-test: robustness to physical perturbations

Subjected to extreme synthetic noise (random masking, baseline shift, time warping), the models reveal their weaknesses. Transformer and BiLSTM maintain the highest resilience against temporal distortions compared to convolutional networks.



XAI: interpreting decisions with Grad-CAM

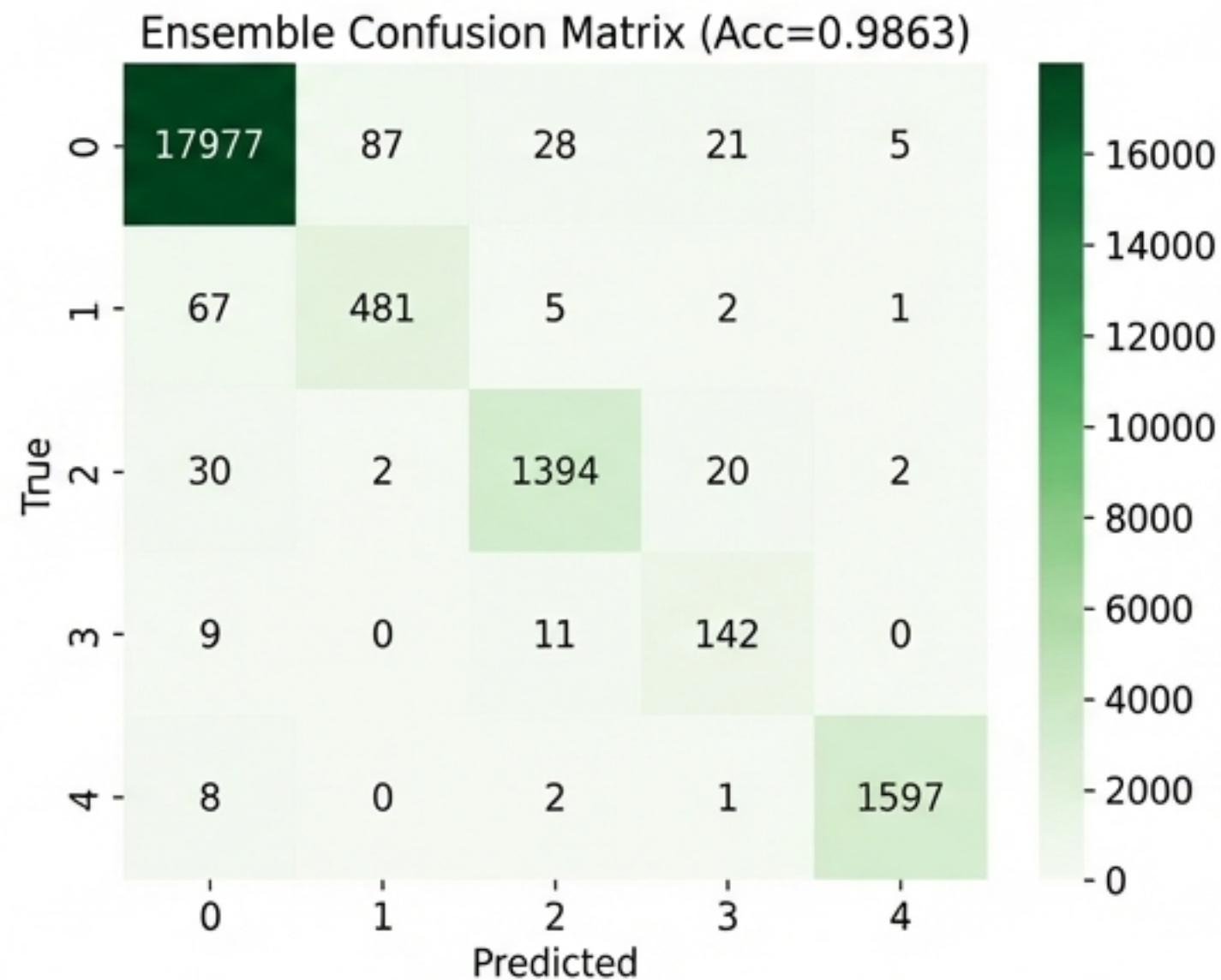
Feature extraction from the last convolutional layer demonstrates the network is not a black box. The model correctly focuses on relevant clinical physiological markers, such as the QRS complex and repolarization anomalies.



The definitive result: weighted ensemble model

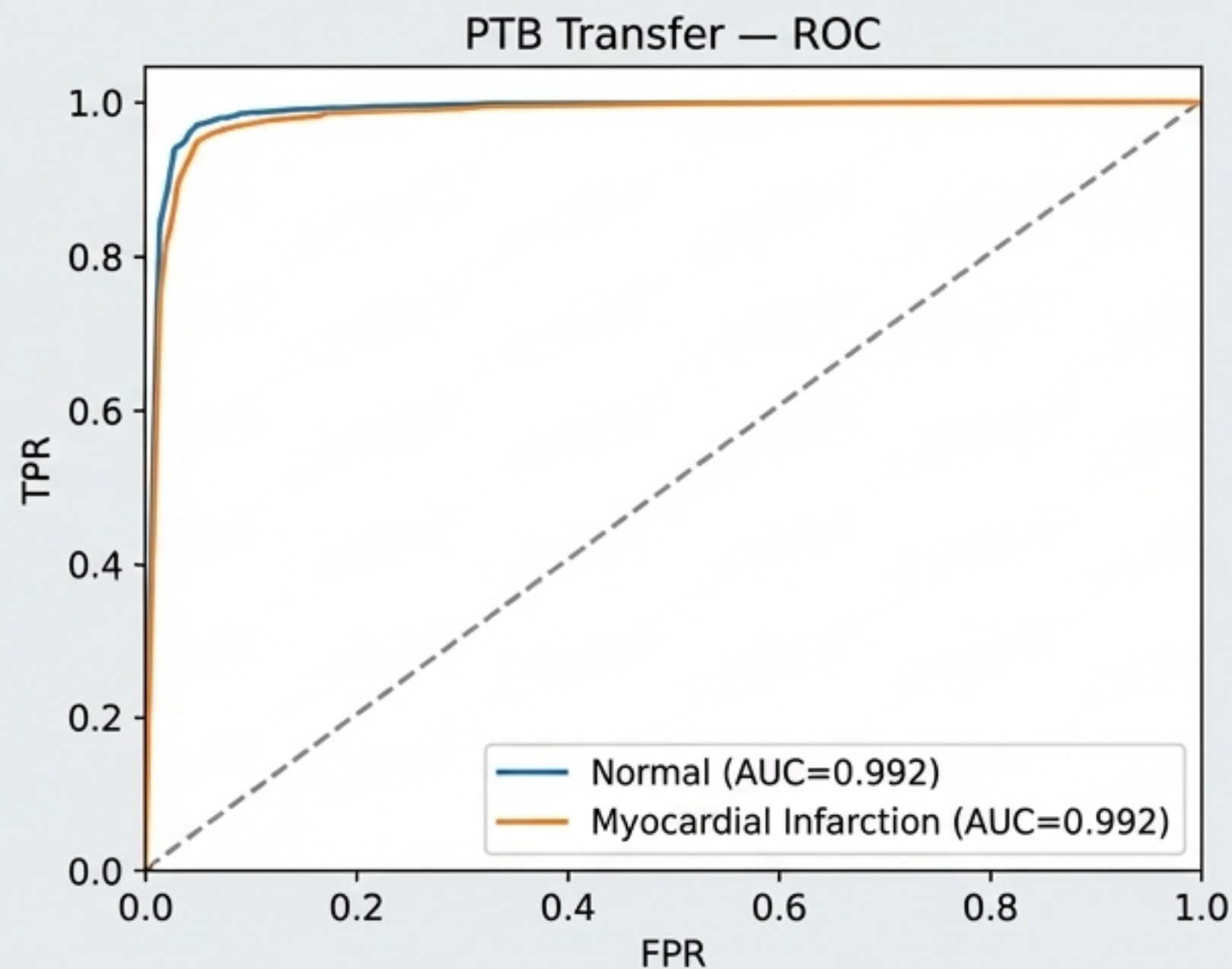
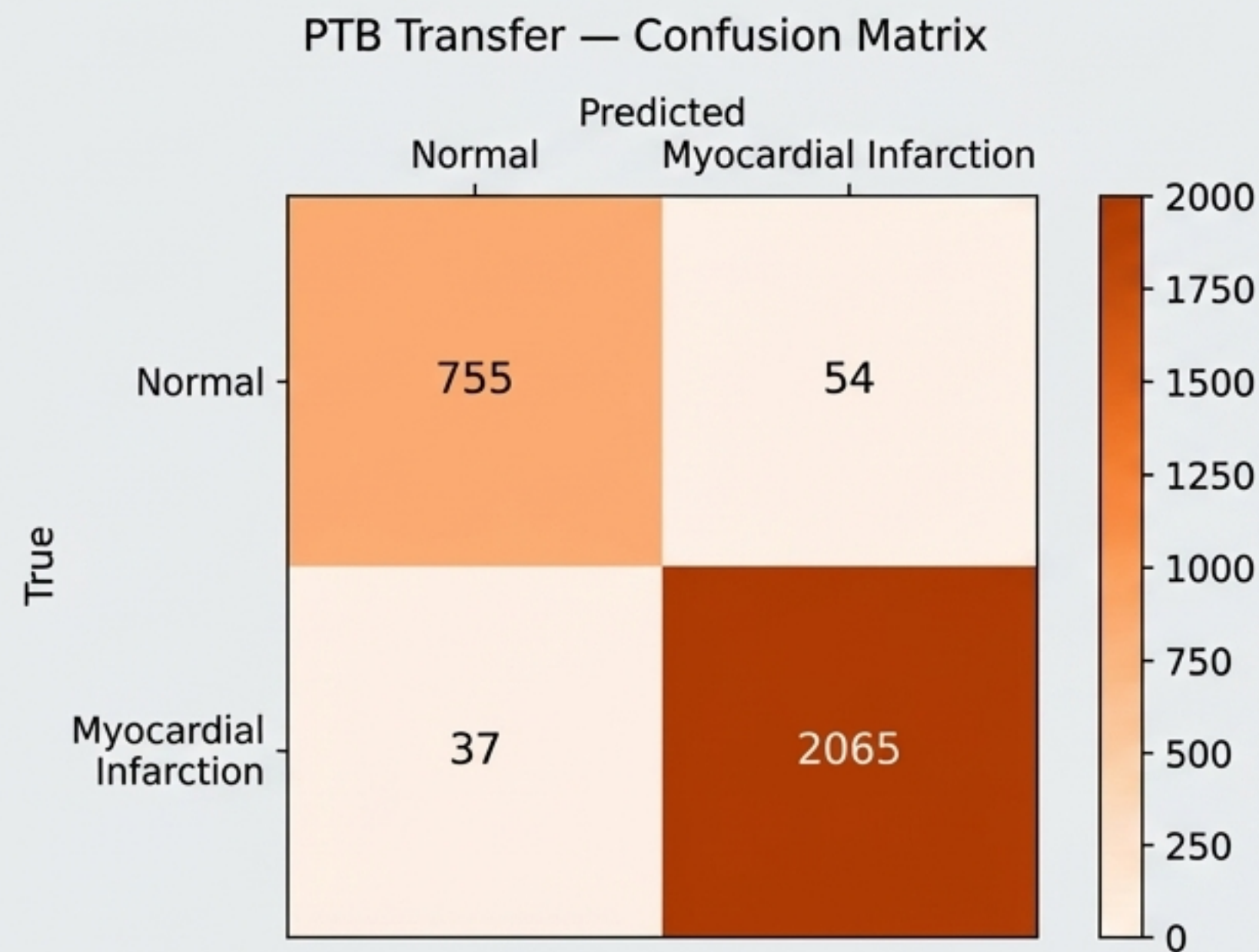
Using a weighted average based on the validation set's macro-F1, the predictions of the 4 models are fused. The ensemble cancels out the biases of individual architectures, reaching a peak performance of 98.63% accuracy and a macro-F1 of 0.9246.

**CNN(w1) +
SE-ResNet(w2) +
BiLSTM(w3) +
Transformer(w4)
= Peak F1**



Transfer learning: generalization on the PTB dataset

The models learned universal features. By freezing the SE-ResNet feature extractor and fine-tuning only the classification head on the PTB dataset (myocardial infarction diagnosis), the system achieves 96.8% accuracy on a completely new domain.



Conclusions and future developments



Rigorous preprocessing: SMOTE and dynamic data augmentation.



Architectural comparison: evaluation between CNN, ResNet, RNN, and Transformers.



Trust & robustness: XAI validation (Grad-CAM) and stress-testing against perturbations.



Generalization: transfer learning demonstrated on parallel diagnostic tasks.



Are you a Data Scientist or HealthTech professional?

Let's discuss ML architectures in the medical domain in the comments.

 GitHub repository available.